AI.407

Robot Navigation

COURSE OVERVIEW

Moving safely through space: global and local path planning, obstacle avoidance, and navigation stacks.



ILLUSTRATIVE HERO IMAGERY

ROBOT NAVIGATION

1

SESSION

60 MINUTES

ORIENTATION: GETTING THERE

KEY TOPICS

- Navigation problem
- Global vs. local navigation
- Navigation stack
- Applications

TEACHING FOCUS

Guide robots from start to goal.

KEY TAKEAWAY Navigation moves robots purposefully.

ROBOT NAVIGATION

2

SESSION

60 MINUTES

GLOBAL PATH PLANNING

KEY TOPICS

- Graph search: Dijkstra, A*
- D* lite
- Roadmaps
- Optimal paths

TEACHING FOCUS

Plan routes over known maps.

▲ **KEY TAKEAWAY** Global planning finds the best route.

3

SESSION

60 MINUTES

LOCAL NAVIGATION & OBSTACLE AVOIDANCE

KEY TOPICS

- Dynamic window approach
- Potential fields
- Vector fields
- Reactive avoidance

TEACHING FOCUS

Avoid obstacles in real time.

KEY TAKEAWAY Local planning reacts to the now.

ROBOT NAVIGATION

4

SESSION

60 MINUTES

NAVIGATION STACKS

KEY TOPICS

- ROS navigation
- Costmaps
- Global and local planners
- Recovery behaviors

TEACHING FOCUS

Assemble a full navigation system.

KEY TAKEAWAY Stacks integrate planning, control, and sensing.

ROBOT NAVIGATION

5

SESSION

60 MINUTES

NAVIGATION IN DYNAMIC WORLDS

KEY TOPICS

- Moving obstacles
- Human-aware navigation
- Social navigation
- Crowd navigation

TEACHING FOCUS

Navigate among people and robots.

KEY TAKEAWAY Social robots move with people.

ROBOT NAVIGATION

6

SESSION

60 MINUTES

ROBOT NAVIGATION & REVIEW

KEY TOPICS

- Navigation demo
- Course recap and assessment
- Toward path planning (AI.408)
- Obstacle course

TEACHING FOCUS

Demonstrate autonomous navigation.

KEY TAKEAWAY A robot that cannot navigate is not autonomous.